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Lesson



Review



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How Does Linkage and Recombination Support DNA Fingerprinting in Forensics?



[Figure 1]

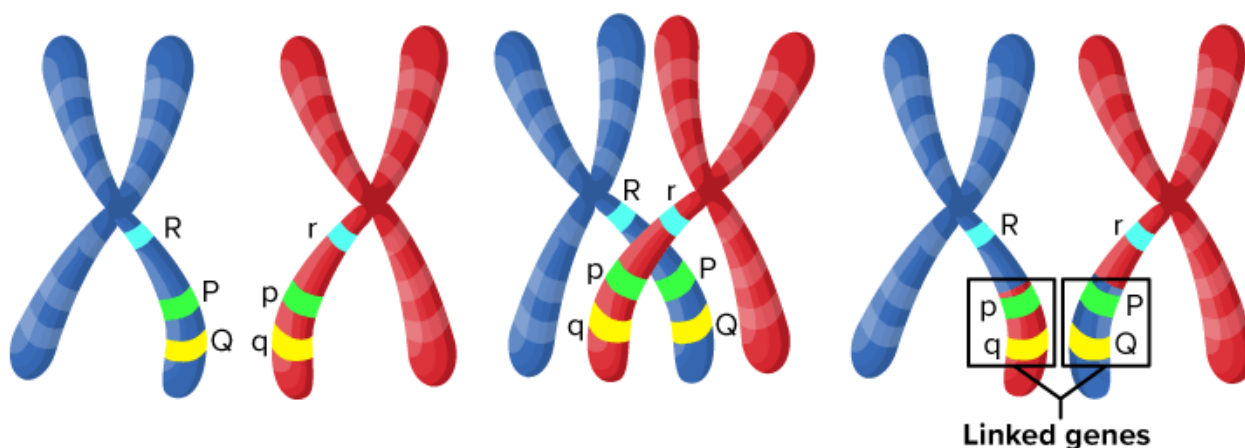
In forensic science, identifying individuals based on their DNA is crucial in solving crimes or identifying victims. Linkage and recombination help scientists understand how genetic markers are inherited together. Certain markers that are closely linked on a chromosome tend to be inherited as a unit. While recombination helps generate variation. By analyzing patterns of linked markers across the genome, forensic experts can match DNA samples with high accuracy. This application relies on the principles of linkage and recombination to establish identity or biological relationships in criminal investigations and paternity cases.

Linkage and Recombination

Linkage describes the phenomenon where **two or more genes that are located on the same chromosome tend to be inherited together**. These genes are called **linked genes**. The closer the genes are to each other physically on the chromosome, the stronger the linkage and the lower the chance they will be separated during **meiosis**.

Unlike Mendel's proposal that all genes assort independently, genetic linkage shows that the proximity of genes on the chromosome can alter inheritance patterns. Thomas Hunt Morgan was the first to demonstrate this phenomenon concretely in his **experiments** with *Drosophila melanogaster*, commonly known as **fruit** flies. He noticed that certain characters, like body colour and eye colour, were often inherited together and did not follow the expected 9:3:3:1 ratio of Mendelian **dihybrid crosses**. He realised this was because these genes were physically located on the **X chromosome** thus linked, preventing their **independent assortment** during meiosis. When genes are very close to each other, they are said to be "**tightly linked**," which reduces the chance that they will be separated by recombination.

Recombination refers to the process by which **new combinations of alleles (gene variants)** are formed in offspring. This happens during meiosis, when **homologous chromosomes** (paired chromosomes from mother and father) exchange segments through a process called **crossing over** (**Figure below**). During this exchange, the genes on the chromosomes are shuffled, producing chromatids that carry new allele combinations that were not present in either parent. This reshuffling of **genetic material** is the key reason recombination greatly increases **genetic diversity** in sexually reproducing **organisms**. Morgan coined the term recombination to describe non-parental (new) gene combinations that result from crossing over.



[Figure 2]

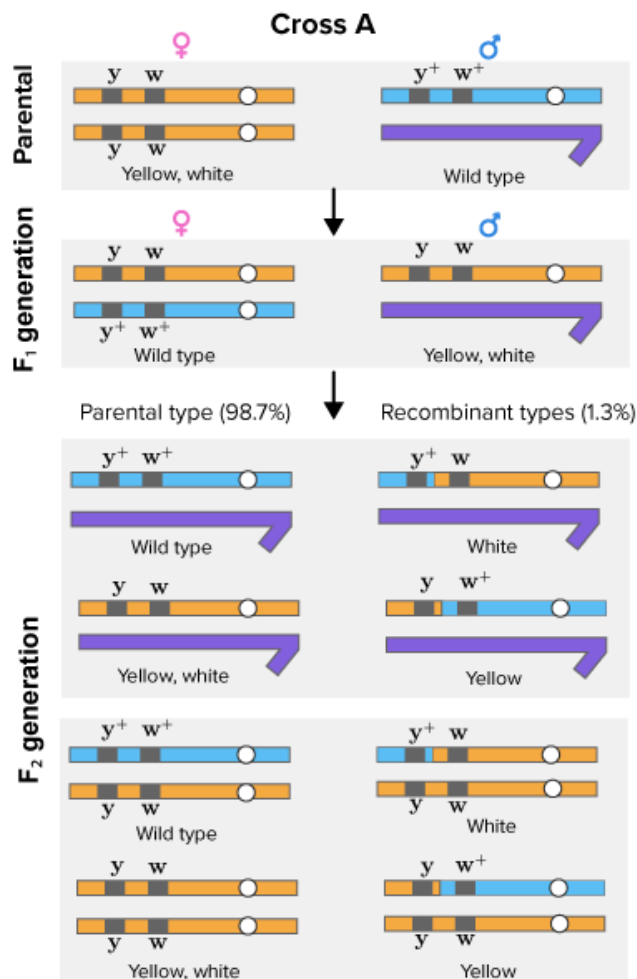
Homologous chromosomes carrying genes (R–P–Q and r–p–q) undergo crossing over during meiosis. The exchange of chromosomal segments can separate linked genes and create recombinant allele combinations, illustrating how the physical distance between genes affects the likelihood of recombination. In this example, genes R and Q are farther apart and therefore undergo crossing over more frequently, producing new combinations (R–p–q and r–P–Q), whereas genes P and Q lie close together and are tightly linked, making recombination between them less likely.

Morgan's Experiments: Unraveling Linkage and Recombination

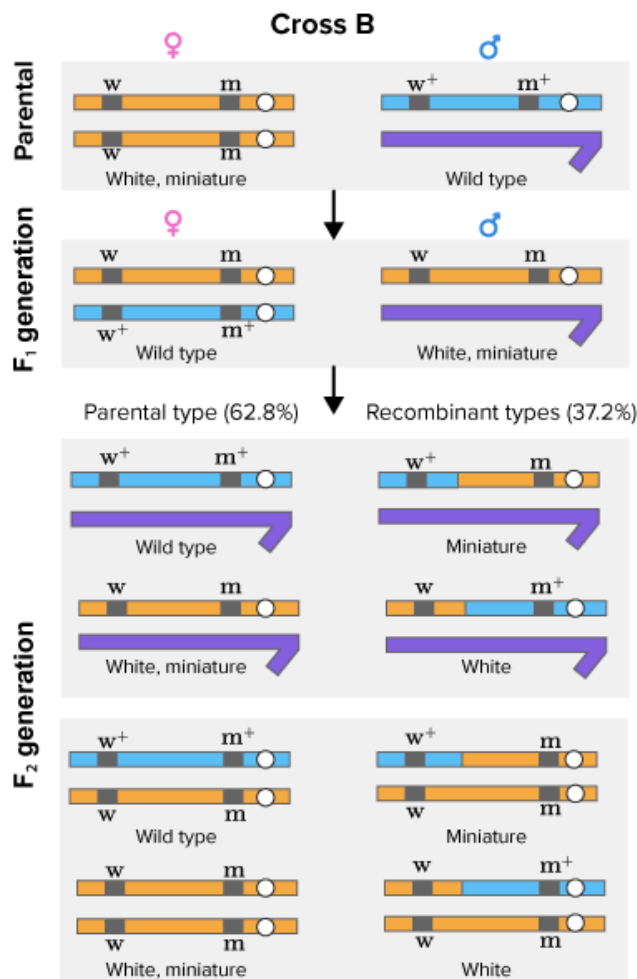
Thomas Hunt Morgan conducted several dihybrid crosses in *Drosophila melanogaster* to study genes located on the X chromosome. These experiments were conceptually similar to Mendel's dihybrid crosses in [pea plants](#), but Morgan observed important deviations from Mendelian expectations. In one such experiment, he crossed yellow-bodied, white-eyed females (both [recessive](#) X-linked characters) with brown-bodied, red-eyed males (dominant, wild-type characters) and intercrossed their F₁ progeny. The resulting F₂ generation did not follow the Mendelian 9:3:3:1 ratio, suggesting that the two genes did not assort independently.

Morgan and his team reasoned that this deviation occurred because the genes controlling body colour and eye colour were located on the same X chromosome and therefore tended to be inherited together. They observed that in such dihybrid crosses, parental gene combinations occurred far more frequently than recombinant (non-parental) types. Morgan explained this physical association of genes on the same chromosome as **linkage**, while the formation of new, non-parental gene combinations resulting from their occasional separation was termed recombination.

Further studies showed that linked genes do not always exhibit the same strength of linkage. Some gene pairs remain associated almost all the time, showing very low recombination and are described as tightly linked, whereas others separate more frequently due to crossing over, indicating loose linkage. For example, Morgan found that the white (*w*) gene, which controls eye colour, and the yellow (*y*) gene, which controls body colour, were tightly linked and showed only 1.3% recombination (Cross A). In contrast, the white (*w*) gene and the miniature (*m*) gene, which controls wing size, were loosely linked and exhibited 37.2% recombination (Cross B).



[Figure 3]



[Figure 4]

Linkage was demonstrated through two dihybrid crosses conducted by Morgan in *Drosophila melanogaster*. Cross A shows recombination between the genes *y* (yellow body colour) and *w* (white eye colour), while Cross B shows recombination between the genes *w* (eye colour) and *m* (miniature wing size). The dominant (wild-type) alleles: y^+ for grey body colour, w^+ for red eyes, and m^+ for normal wing size, are indicated by a superscript plus sign (+). The frequency of recombinant offspring is much lower in Cross A (1.3%) than in Cross B (37.2%), indicating that the linkage between *y* and *w* is stronger than the linkage between *w* and *m*.

***Drosophila*: The Model Organism for Genetics**

Drosophila melanogaster has been a favourite among geneticists due to its rapid life cycle, ease in breeding and handling, clear observable traits, and simple chromosome structure. Morgan's work with *Drosophila* revolutionized biological understanding of inheritance patterns. In schools and universities, *Drosophila* remains a prominent model organism for hands-on genetics experiments, showing students real examples of linkage and recombination. For instance, students may observe how traits like eye colour and wing shape are inherited together in fruit fly crosses.

Video: Thomas Morgan & Fruit flies



Genetics - Thomas Morgan & Fruit flies

Sri Chaitanya Academy NEET



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Sex-Linked Genes and their Inheritance

Many of the genes Morgan studied, such as those governing body colour and eye colour in *Drosophila*, were located on the X chromosome, making them "sex-linked." Sex-linked genes exhibit unique patterns of inheritance because males (XY) and females (XX) have different number of X chromosomes. Because of this, traits governed by sex-linked genes can appear more frequently in one sex than the other, and the linkage between sex-linked genes can further influence these patterns. For example, haemophilia and red-green colour blindness in humans are classic examples of X-linked inheritance.

Males are hemizygous for X-linked genes because they possess only one X chromosome and therefore only one copy of each X-linked gene. As a result, any recessive allele present on the

single X chromosome is expressed, since there is no corresponding allele to mask its effect. For example, a male fly inheriting the white-eye allele on its X chromosome will display white [eyes](#), even though the allele is recessive.

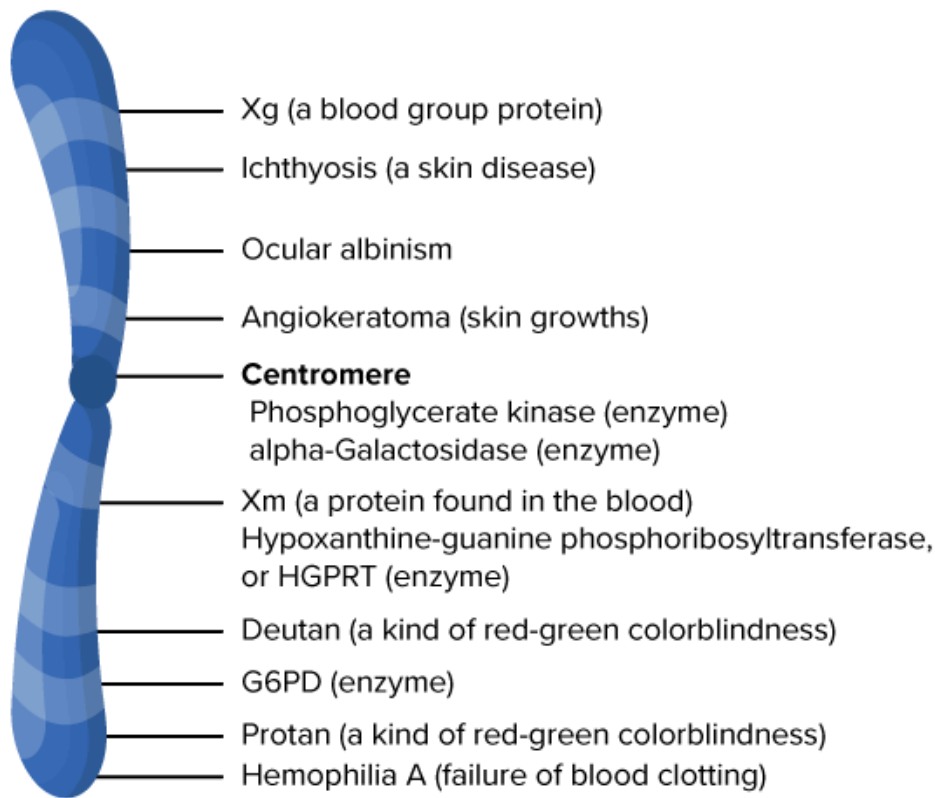
Females, having two X chromosomes, can be homozygous or heterozygous for an X-linked gene. A heterozygous female carries one normal (wild-type) allele and one white-eye allele and is therefore described as a carrier. She usually shows the wild-type [phenotype](#) but can transmit the white-eye allele to her offspring. A female will express the recessive white-eye trait only if she inherits two copies of the white-eye allele.

Gene Mapping: Using Recombination to Locate Genes

Morgan's student, Alfred Sturtevant, realized the frequency of recombination between two genes could be used to estimate the distance between them on a chromosome. Higher recombination frequencies meant the genes were further apart; lower frequencies indicated proximity. Sturtevant constructed the first "genetic map" by plotting gene positions based on recombination percentages. These maps, essential to modern genetics, were vital stepping stones for complex projects like the [Human Genome Project](#). In recent years, gene mapping has helped locate genes linked to inherited diseases, improving diagnostics and treatments.

Genetic maps, or linkage maps, visually represent the relative positions of genes on a chromosome, based on recombination frequencies. The unit of measurement on these maps is the "map unit" or "centimorgan (cM)," where 1% recombination equals one map unit. Frequencies of crossing over can be used to construct a [linkage map](#) as shown in the **Figure below**. A linkage map shows the locations of genes on a chromosome.

Genetic maps are foundational tools in genomics, guiding researchers in identifying the locations of genes. For example, in Indian [agriculture](#), genetic maps help scientists identify markers for disease resistance or high yield, enabling the [development](#) of improved crop varieties through marker-assisted selection techniques.



[Figure 5]

Linkage Map for the Human X Chromosome. This linkage map shows the locations of several genes on the X chromosome. Some of the genes code for normal proteins. Others code for proteins that are associated with genetic disorders. Which pair of genes would you expect to have a lower frequency of crossing over: the genes that code for hemophilia A and G6PD deficiency, or the genes that code for Protan and Xm?

Significance of Linkage and Recombination

- **Genetic Mapping:** Recombination frequencies help scientists create genetic linkage maps, which show the relative positions of genes on chromosomes.
- **Crop Improvement:** Plant breeders use linkage and recombination knowledge to combine beneficial traits, such as pest resistance and high yield, into a single plant variety.
- **Medical Genetics:** Linkage analysis is used to trace the inheritance of disease genes in families, helping in genetic counselling and disease **prediction**.
- **Forensics and DNA Fingerprinting:** Linked DNA markers are used to match individuals at crime scenes.

Video: Gene Linkage and Genetic Maps



Gene Linkage and Genetic Maps

Professor Dave Explains



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Summary of Linkage and Recombination

- Linkage refers to the tendency of genes located on the same chromosome to be inherited together. The closer two genes are, the stronger the linkage and the lower the chance of separation by crossing over.
- Morgan's *Drosophila* experiments revealed that linked genes do not follow Mendelian 9:3:3:1 ratios, as parental combinations appear far more frequently than recombinants.
- Recombination occurs during meiosis when homologous chromosomes exchange segments, shuffling alleles and producing new allele combinations not found in either parent.
- The recombination frequency between two genes indicates their distance on a chromosome; low recombination means tight linkage, while high recombination means genes are farther apart.
- Morgan's group found examples of both tightly linked genes (e.g., y and w, 1.3% recombination) and loosely linked genes (e.g., w and m, 37.2% recombination).
- Alfred Sturtevant used recombination frequencies to create the first genetic linkage map, laying the foundation for modern genome mapping.
- *Drosophila melanogaster* became a key model organism because of its short life cycle, easily observable traits, and well-defined chromosomes.
- Sex-linked genes, often located on the X chromosome, show unique inheritance patterns because males are hemizygous for X-linked genes; recessive traits appear more frequently in males.
- Linkage and recombination are central to applications such as genetic mapping, crop improvement, medical genetics, and forensic DNA analysis.

Review Questions on Linkage and Recombination

1. What is linkage? How does the distance between genes affect the strength of linkage?
2. Define recombination. During which process does it occur?
3. State three advantages of using *Drosophila* as a model organism for genetic studies compared with Mendel's pea plants.
4. Why did Morgan's dihybrid crosses in *Drosophila* deviate from Mendel's expected 9:3:3:1 ratio?
5. What recombination frequencies did Morgan observe between the gene pairs y–w and w–m? What did these frequencies indicate?
6. Explain how recombination produces allele combinations that were not present in either parent.
7. Who constructed the first genetic linkage map, and what principle was it based on?

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